TESTS ABOUT THE POPULATION MEAN

Exercise 1: One sample t-test

The aim of this exercise is to use SPSS to carry out the one-sample t-test, including checking it's assumption, and to interpret the results.

A research study was conducted to find out whether the mean birth weight of the babies in this sample are equal to a population level birth weight of 2500 g (use the dataset: *lowbirthweight.sav*).

Research Question: Does the mean birth weight of babies = 2500g?

Step 1: Check the assumption for a one-sample t-test, i.e., are the birth weights (*bwt*) normally distributed in the sample? (Hint: Use the chartbuilder to produce a histogram).

Step 2: State the Null Hypothesis and the Alternative Hypothesis.

Step 3: Find the mean and SD of birth weight.

Step 4: Perform the one-sample t-test (Analyze/One-Sample T-Test) and put the *bwt* in the Test variable and the population value of 2500 in the Test value.

Step 5: Find the mean difference between the sample and population and its 95% CI. Record the p-value and interpret the results (including size and direction).

Exercise 2: Independent samples t-test

The aim of this exercise is to carry out the independent samples t-test, including checking it's assumptions, and to interpret the results.

A research study was carried out to determine whether there is a difference in total cholesterol between males and females (data source: *cardiacdata.sav*) Research Question: Is the mean total cholesterol (*tchol*) of males and females different?

Step 1: Check the assumption for the independent samples t-test, i.e., is total cholesterol (*tchol*) normally distributed in each of the groups? (Hint: Use the chartbuilder to produce a histogram and use Groups/Point ID to produce separate histograms for males and females).

Step 2: State the Null Hypothesis and the Alternative Hypothesis.

Step 3: Find the mean and SD of total cholesterol for males and females separately.

Step 4: Perform the independent samples t-test (Analyze/compare means/Independent-Samples T-Test) and put *tchol* in the Test variable and sex (1 for female 2 for male) in the Grouping variable.

Step 5: Check the assumption of equal variances using the p-value of Levene's test.

Step 6: Based on Levene's test, find the appropriate row of the t-test results and use it to interpret the findings (using the mean difference, it's 95% CI, and the p-value).

Exercise 3: Paired t-test

A study is conducted to find out whether baby weight is different between soon after delivery (birth weight) and at 3 months (Data source: *lowbirthweight.sav*)

Research Question: Is there a change in baby weight between baseline and 3 months?

Step 1: First find the difference between baby weight at birth and at 3 months (*diffbwt=mth3wt-bwt*) for each individual and check the assumption for the paired t-test, i.e., are the birth weight differences (*diffbwt*) normally distributed in the sample? (Hint: Use the chartbuilder to produce a histogram).

Step 2: State the Null Hypothesis and the Alternative Hypothesis.

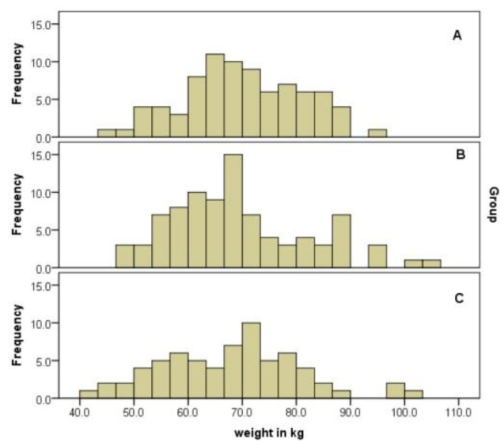
Step 3: Find the mean and SD of the difference in birth weights.

Step 4: Select the paired t-test (Analyze/Paired-Samples T-Test) and enter into the 'Paired Variables' box *mth3wt* under variable 1 and *bwt* under variable 2.

Step 5: Find the mean difference in baby weight and its SD. Record the p-value and 95% CI of the mean difference and interpret the results.

Exercise 4: ANOVA

Scenario: A student trying to get some practice in carrying out ANOVA in SPSS decided to compare weights (kg) between three independent groups (A, B & C) with a null hypothesis (H_0) that there is no difference in mean weight between groups in the population. The student first plotted histograms of weight for the 3 groups.



1. Comment on the distribution of weight in the 3 groups. Which assumption is being checked here?

2. The student then obtained the following descriptives:

Descriptives

weight in kg

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
A	81	70.1341	11.03751	1.22639	67.6935	72.5747	44.91	93.44
B	88	69.2822	13.03126	1.38914	66.5211	72.0432	46.72	105.24
C	67	68.0064	12.87219	1.57259	64.8666	71.1462	42.18	101.61
Total	236	69.2124	12.31197	.80144	67.6334	70.7913	42.18	105.24

Percentiles

Group			Percentiles 25	50	75
Weighted Average(Definition 1)	weight in kg	A	62.8250	69.8500	78.2450
		B	60.3300	66.6800	78.5850
		C	58.0600	69.8500	75.3000
Tukey's Hinges	weight in kg	A	63.0500	69.8500	78.0200
		B	60.3300	66.6800	77.7900
		C	58.2850	69.8500	75.0700

a) What was the sample size? Comment on the size of the sample with respect to carrying out ANOVA.

b) Which descriptives should be presented? Write them out

3. The student followed the procedure on conducting an ANOVA in SPSS but could not quite remember what one of the tables in the output window was for.

weight in kg

Levene Statistic	df1	df2	Sig.
.774	2	233	.462

a) What is the purpose of the test whose result is shown in the table above?

b) Interpret the results of the table.

ANOVA

weight in kg

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	166.681	2	83.340	.548	.579
Within Groups	35455.703	233	152.170		
Total	35622.384	235			

4. The student decided that the above table showed evidence that there is a difference in weight between groups

a) Is this statement correct?

b) Interpret in full the results of the table.

5. The student then decided to do a post-hoc test as follows:

Multiple Comparisons

Dependent Variable: weight in kg

Scheffe

(I) Group	(J) Group	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	Upper Bound
A	B	.85191	1.89944	.904	-3.8275	5.5313
	C	2.12766	2.03712	.580	-2.8909	7.1462
B	A	-.85191	1.89944	.904	-5.5313	3.8275
	C	1.27574	2.00010	.816	-3.6516	6.2031
C	A	-2.12766	2.03712	.580	-7.1462	2.8909
	B	-1.27574	2.00010	.816	-6.2031	3.6516

a) What is your comment on this next step of the analysis?

b) Was this an appropriate step bearing in mind the ANOVA results?

7

Comparison of Means

Nancy Berman

Summary

This chapter describes statistical methods to test for differences between means or other measures of central tendency of 2 or more populations. Parametric tests and nonparametric tests are included. Methods for pairwise comparisons when more than 2 groups are being compared are included.

Key Words: Analysis of variance; contrasts; Kruskal-Wallace test; *t*-test; Wilcoxon-Mann-Whitney test; Wilcoxon signed rank test.

1. Introduction

In this chapter, we will present methods to test hypotheses about means. We will use Student's *t*-test to test hypotheses for paired samples or to compare 2 independent groups and we will use analysis of variance (ANOVA) to compare means in more than 2 independent groups. We will also describe tests that may be used when the *t*-test or ANOVA would not be appropriate. These are called nonparametric tests, which compare measures of central tendency that are different from the mean. In addition, we will introduce the *F* distribution, which is used in ANOVA and also in comparing variances of independent groups.

We will use the following notation:

Parameter	Population value	Sample value
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Variance	σ^2	s^2
Sample size		n
Sample value		x_i

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When there is more than one sample to discuss, we will use numeric subscripts to indicate the groups (e.g., μ_1, $\bar{x}_1$, s_1, n_1 for group 1; μ_2, $\bar{x}_2$, m_2, s_2, and n_2 for group 2, etc.). We will use double subscripts for values in each group (e.g., x_{ij}, where i is the group and j is the sample number). Other notations will be introduced as needed.

2. Test Statistics

2.1. The t-Test

In **Chapter 4**, you were introduced to the t-test for testing the hypothesis that the mean of a sample was equal to a given value. In this chapter, we will show the use of the t-test for two other comparisons of means, first when the samples are paired and second for independent groups. We recall that the t-statistic has the general form

$$t = \frac{\text{difference of means}}{\text{standard error}},$$

where the standard error is the standard error of the numerator. The t-distribution is a "sampling" distribution, that is, it does not assume any knowledge about the true parameters. It is important to remember that the t-distribution is actually a family of distributions, each one identified by a parameter called the *degrees of freedom* (d.f.), which is related to the sample size. It is similar in shape to the standard normal distribution, and as the degrees of freedom increase, it becomes more similar (**Fig. 1**).

If you have computed a t-statistic and are looking it up in a table to calculate the P value, you will need to use the degrees of freedom to get the correct P value. If you are using a computer program, it will probably compute this for you from the sample size.

2.2. The F Distribution

The F statistic is another sampling distribution that will be used in this chapter. The F statistic is the ratio of two independent random variables that have chi-square distributions (**Chapter 5**). In this chapter and in many other applications, the F statistic is the ratio of 2 independent sample variances:

$$F = \frac{s_1^2}{s_2^2}. \tag{1}$$

Note that there are two parameters that go with an F distribution, the degrees of freedom for the numerator and the degrees of freedom for the denominator.

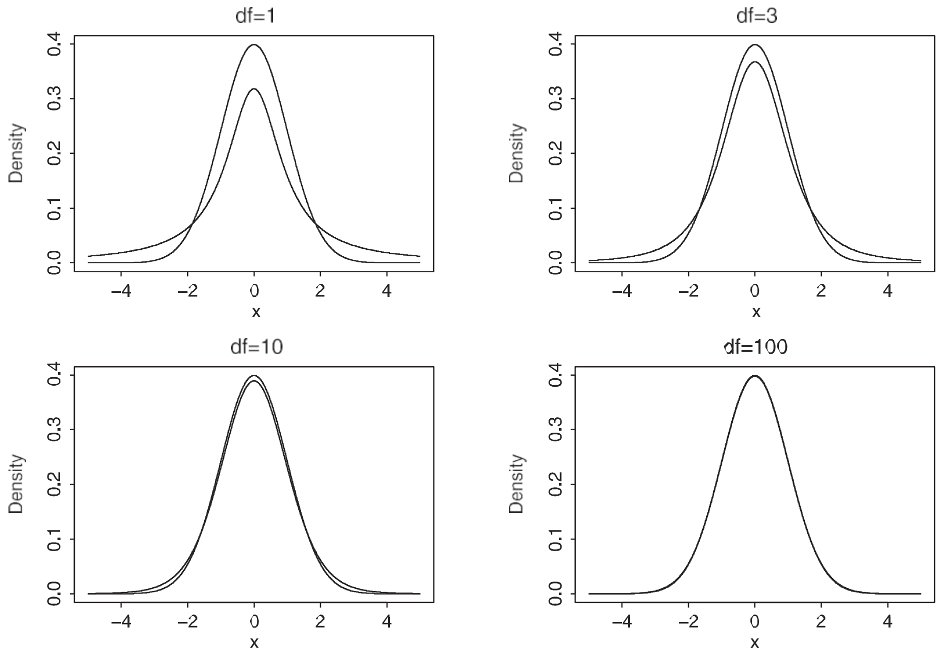

Fig. 1. Values of the t-distribution for different degrees of freedom.

3. *t*-Tests Comparing Two Means

3.1. Paired Samples

Paired samples means that there are two groups of samples, that each sample in the first group is related to one and only one sample in the second group, and that there are no unpaired samples in either group. Both groups have the same sample size, n. The most common types of paired samples are measurements on the same subject before and after an intervention, measurements on a group of experimental animals and their pair-fed matches, or measurements on subjects from an affected group and matched subjects from a control group. In order to compute the t-statistic, you must first compute the difference between each of the paired values,

$$d_i = x_{i1} - x_{i2}, \tag{2}$$

and then compute the mean and standard deviation, $\bar{x}_d$ and s_d, of the differences. It is assumed that the d_i have a normal distribution. This is guaranteed if both groups are assumed to come from underlying normal distributions.

Usually the difference is compared with zero, so the null and alternate hypotheses are

$$H_0 : \delta = 0$$

$$H_1 : \delta \neq 0,$$

where δ is the (unknown) true value of the mean difference.

The *t*-statistic is simply

$$t = \frac{\bar{x}_d - c}{s_d / \sqrt{n}}. \tag{3}$$

If you have detected that this looks a lot like the *t*-test on a single sample, you are right! There are a few points to remember. One is that although the mean of the difference, $\bar{x}_d$, is equal to the difference of the means, $\bar{x}_1 - \bar{x}_2$, you can't compute the standard deviation for the difference from the standard deviations of the 2 groups; you must compute the differences and then their standard deviation. Another is that you can call either group 1 or 2, but you usually do this based on your hypothesis. If you want to compare the difference to a constant, *c*, other than zero, and your software doesn't support it, then you can subtract *c* from the values or mean in the first group, but not in both.

Example 1

Table 1 shows pre- and posttreatment data from a sample of subjects who participated in a study of the psychobiology of depression ***(1)***. The sample value is cerebrospinal fluid: 3-methoxy-4-hydroxyphenylglycol (CSF MHPG). The groups are composed of unipolar depressed subjects and bipolar depressed subjects. We include the difference between values before and after treatment. We will test the null hypothesis that the difference is not zero in the unipolar group.

Using **Equation 3**, the *t*-statistic for the unipolar group is

$$t = \frac{31.61}{8.20/\sqrt{12}} = 13.35 \quad \text{with 11 degrees of freedom.}$$

The *P* value for this statistic is <0.0001, so we conclude that there is a significant change in CSH MHPG in the unipolar group.

We can test the same hypothesis for the bipolar group. The *t*-statistic is

$$t = \frac{16.33}{6.89/\sqrt{12}} = 8.21 \quad \text{with 11 degrees of freedom.}$$

The *P* value for this statistic is also <0.0001. Therefore, we conclude that there is also a significant change in CSH MHPG in the bipolar group.

In the next section, we will use a *t*-test to see if there are differences between the groups in the amount of change.

Table 1
Pre- and Posttreatment Levels and Change in CSF MHPG for 2 Diagnostic Groups (expressed in pmol/ml)

	Unipolar			Bipolar		
	Pretreatment	Posttreatment	Change	Pretreatment	Posttreatment	Change
	63.7	36.7	27	26.5	18.3	8.2
	61	37.8	23.2	38.6	25.8	12.8
	59	32.8	26.2	59	24.6	34.4
	65.2	24.6	40.6	42.8	28.4	14.4
	59.6	24.4	35.2	28.7	19.2	9.5
	59.7	22.7	37	47.7	32.6	15.1
	79.3	38.5	40.8	44.7	27.9	16.8
	60.6	28.9	31.7	47.7	28.6	19.1
	69.5	32.7	36.8	52.7	35.3	17.4
	54.9	37.7	17.2	54.6	37.8	16.8
	54.2	31.8	22.4	40.8	19.8	21
	67.5	26.3	41.2	40.8	30.3	10.5
Mean	62.85	31.24	31.61	43.72	27.38	16.33
Var	47.68	33.26	67.26	93.76	38.81	47.46
SD	6.91	5.77	8.20	9.68	6.23	6.89
Median			33.45			15.95
n	12	12	12	12	12	12

3.2. The Two-Sample t-Test in Independent Groups

The t-test is also used to compare means in 2 independent groups. In this case, the groups are independent, there is no matching, and the sample sizes may be different. There are 2 assumptions. The first is that each sample has an underlying normal distribution. This may be relaxed somewhat if the variables are continuous and the distribution is symmetric. The other assumption is that the standard deviations in the 2 samples are the same. This is more restrictive, and we will address what to do if this is violated later in this section. For now, we will assume that both assumptions are met. The null and alternate hypotheses are

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2.$$

The quantity we are interested in is the difference between the population means, $\mu_1 - \mu_2$. We use both sample standard deviations to compute the "pooled" standard deviation as

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}. \quad (4)$$

The t-statistic comparing the means is simply

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \tag{5}$$

which has $n_1 + n_2 - 2$ degrees of freedom.

You will recall that one of the assumptions for the t-test on independent samples is that they have the same variance. If the data in 2 groups are normally distributed, this assumption may be tested using an F statistic. The hypotheses are

$$H_0 : \sigma_1^2 = \sigma_2^2$$

$$H_1 : \sigma_1^2 \neq \sigma_2^2.$$

The F statistic to test this hypothesis is the ratio of the sample variances (see **Section 2.1**), with the larger variance in the numerator:

$$F = \frac{s_1^2}{s_2^2}. \tag{6}$$

The degrees of freedom for the numerator are $n_1 - 1$ and the degrees of freedom for the denominator are $n_2 - 1$. If F statistic is very large, so that the P value is small, then there is evidence that the variances are not equal.

The F statistic is most widely used for testing equality of variances for 2 independent samples, however it is not always accurate when the distribution of the two samples is not precisely normal. An alternative is the Brown-Forsythe (BF) test ***(2)***, which looks at the spread of the data in absolute terms. It is usually used with comparison of multiple groups in ANOVA (see below) but may be used with 2 groups. To compute the BF statistic, we first calculate the median of each group, md_i, and then compute the absolute value of the difference between each sample and its group median:

$$bf_{ij} = |x_{ij} - md_i|. \tag{7}$$

The BF test is simply a 2-sample t-test comparing the bf_{ij} in the two groups.

If the variances of the 2 groups are not equal, a modified estimate of the standard error of the difference is used and the degrees of freedom for the t-statistic are adjusted. The standard error of the difference is estimated as

$$s_{np} = \sqrt{\frac{s_1^2 + s_2^2}{n_1 + n_2}} \tag{8}$$

and the t-statistic is

$$t' = \frac{\bar{x}_1 - \bar{x}_2}{s_{np}}. \tag{9}$$

The distribution of this statistic can be approximated by a t-distribution with a different number of degrees of freedom. Several statisticians have developed methods for estimating the best matching distribution; the one most commonly used at this time is due to Satterthwaite ***(3)***. The formula for computing these degrees of freedom is complicated and will not be given here.

Example 2

In **Example 1**, we computed the change after treatment in CSF MHPG in both the unipolar and bipolar groups (**Table 1**). We would like to test whether the change is the same in both groups. The null hypothesis is that the mean change in unipolar and bipolar groups are equal. To test whether variances are equal in the 2 groups, using **Equation 6**, the F statistic is

$$F = \frac{67.26}{47.46} = 1.42.$$

The n for the unipolar group is 12, the n for the bipolar group is also 12, so that the F statistic has 11 and 11 degrees of freedom. From a table, the P value for an F statistic with value 1.42 and 11 and 11 degrees of freedom is 0.57, so we can assume that the variances are equal.

To use the BF test, the absolute differences from the medians for the changes in CSH MHPG in **Table 1** are shown in **Table 2**. The t-statistic for these data is 1.28 with 22 degrees of freedom, which has a P value of 0.2131. This test also does not reject the null hypothesis that the variances are equal.

Therefore we compute the test statistic using **Equation 4** and **Equation 5**:

$$s_p = \sqrt{\frac{11 \cdot 67.26 + 11 \cdot 7.46}{22}} = 7.57$$

and

$$t = \frac{31.61 - 16.33}{7.57 \cdot \sqrt{\frac{1}{12} + \frac{1}{12}}} = 4.94 \quad \text{with 22 degrees of freedom.}$$

The P value for this statistic is <0.0001 so we can conclude that there is a difference between the 2 groups in the amount of change.

A good software package may automatically perform the F test and give you the results of the F test along with the t-value, degrees of freedom, and P value

Table 2
Absolute Deviation from the Median

	Unipolar	Bipolar
	6.45	7.75
	10.25	3.15
	7.25	18.45
	7.15	1.55
	1.75	6.45
	3.55	0.85
	7.35	0.85
	1.75	3.15
	3.35	1.45
	16.25	0.85
	11.05	5.05
	7.75	5.45
Mean	6.99	4.58
Var	17.63	24.70
SD	4.20	4.97
n	12	12

for both equal and unequal variances. Other packages may allow you to do the *F* test separately and then choose the *t*-test that goes with the results. Most packages do not do the BF test as a *t*-test option, but you may be able to do it by computing the deviations and using a *t*-test. Alternatively, the *t*-test on independent groups is equivalent to an ANOVA on 2 groups (**Section 5.1**), so you may use ANOVA software to test this. If your software assumes equal variances and does not allow for adjusted *t*-values, then, unless you can be sure that the variances are equal, you should try to run it on a different package or use a nonparametric approach (**Section 4**).

4. Tests of Central Tendency When the Distribution Is Not Normal

The basic assumption of a *t*-test is that the underlying distribution of the group(s) is normal. It is a fairly robust test, so that it may be appropriate if the sample size is large and the distribution is symmetric about the central value. Sometimes the data can be transformed so that it meets the assumption of a normal distribution. For example, many biological variables have a distinct positive skew. Often, the logarithm of the variables has a normal distribution, so that you can compare the samples using the log-transformed values. Just remember that you are comparing the geometric mean, not the arithmetic mean when you compare log-transformed variables.

Table 3
Twelve Numbers with Ranks

Value	2	19	23	31	35	47	56	56	59	61	98	98
Rank	1	2	3	4	5	6	7.5	7.5	9	10	11	12

If the data cannot be transformed to have a normal distribution, then the *t*-test will not give the correct *P* value. There are alternatives to the *t*-test for non-normally distributed variables. These tests are referred to as nonparametric tests. They are usually based on ranks rather than actual values. To compute the ranks for a sample, the values are arranged in increasing order and then numbered sequentially from 1 to the sample size, *n*. Sometimes 2 or more samples will have the same value. In this case, each one gets the average of the 2 or more ranks that are tied. **Table 3** shows the ranking for 12 numbers with 1 tie. These tests also use the median as the measure of central tendency, rather than the mean.

Most software packages can compute nonparametric tests and give the *P* values for test statistic. Alternatively, the *P* values may be given in tables. Many tables for nonparametric tests just give the critical values for certain sample sizes and other statistics. If the sample size is large enough, >20, then there is a large sample statistic that can be used that has a distribution close to a standard distribution, such as the normal or chi-square distribution. Most software packages compute this large-sample statistic and *P* value, as well as or instead of the exact statistic.

4.1. The Sign Test for a Single Sample

The null hypothesis for this test is that the median of the sample is a certain value, *C*. Using θ to represent the true value of the median this is stated as

$$H_0 : \theta = C$$

$$H_1 : \theta \neq C.$$

The sign test is based on the binomial test (**Chapter 5**). The data is arranged in order and the number of samples that are bigger than the median is computed. If the median is *C*, then you would expect half the samples to be larger than *C* and half to be smaller, so that $P(>C) = P(<C) = 0.5$. Suppose you have a sample size of *n* and *r* samples are larger than *C*; then using the binomial distribution the probability of getting *r* samples larger than *C* is

$$p(r) = \frac{n!}{r!(n-r)!} 0.5^r (1-0.5)^{n-r} = \frac{n!}{r!(n-r)!} \left(\frac{1}{2}\right)^r. \tag{10}$$

Table 4
Mood Scores in 12 Unipolar Subjects

3.45	4.56	4.68	5.06	5.41	6.63	6.99	7.81	8.22	8.81	9.14	9.86

Many statistical books have tables of the binomial probability for different combinations of n and r. Because hypothesis testing is based on the probability of getting the observed result or something rarer for a 2-sided test, we must compute the binomial probability for $r + 1$, $r + 2$, and so on, up to n. Then these are added together to give the answer (i.e., the probability of getting this number of samples larger than C if the median value is C).

Example 3

Table 4 shows the behavioral scores for the same group of unipolar depressed subjects that were in **Table 1**. We assume from the literature that the true median score is 5. There are 9 samples larger than C.

Using the binomial distribution the probability of getting 9 or more samples is computed as in **Table 5**.

For a 2-sided test, we calculate $2 \times 0.0730 = 0.1460$ and we do not reject the null hypothesis that the median is C.

For large samples, $n > 20$, the following has approximately a standard normal distribution:

$$z = \frac{X - 0.5n}{0.5\sqrt{n}}, \tag{11}$$

where $X = r - 0.5$ if $r \geq n/2$ or $X = r + 0.5$ if $r < n/2$.

Example 3 (Continued)

We illustrate using the large sample approximation (although the exact distribution is more appropriate for this example). We let $X = 9 - 0.5 = 8.5$, then

Table 5
Binomial Probabilities for n = 21 and $r \geq 9$

	r	$P(r)$
	9	0.0537
	10	0.0161
	11	0.0029
	12	0.0002
Sum		0.0730

$z = 1.44$ and the 2-sided P value is 0.150, which agrees closely with the exact distribution.

4.2. The Wilcoxon Signed Rank Test for Paired Samples

The most commonly used nonparametric test to compare paired samples is the Wilcoxon signed rank test. This is similar to the sign test but has the advantage that the magnitudes of the ranks are included in the calculations. It assumes that the differences are distributed symmetrically about the median difference, which would be expected if there were no real underlying difference between the paired values. As with the paired t-test, the analysis is based on the difference between the paired values. Using θ to represent the median of the differences, the hypotheses are

$$H_0 : \theta = 0$$

$$H_1 : \theta \neq 0.$$

The process is as follows. The differences are ranked on the basis of their magnitude, ignoring the sign. Then the signs are restored to the rankings. The sum of the positive ranks, S_p, and of the negative ranks, S_n, are computed. The test criterion, T, is the smaller of these 2 sums. If the sample is small (<16), then the critical value for T must be looked up in a table ***(4)***. If the sample is large, then you may use the statistic in **Equation 12**, which has an approximately standard normal distribution:

$$z = \frac{T + 0.5 - 0.25n(n+1)}{\sqrt{n(n+1)(2n+1)/24}}. \tag{12}$$

Example 4

Table 6 shows the behavioral scales for the unipolar depressed subjects before and after treatment. The third column shows the differences, the fourth column shows the ranks on unsigned values, and the fifth column shows the signed ranks.

For this example, the sum of the positive ranks is 57 and the sum of the negative ranks is 21, so $T = 21$. Tables ***(4)*** show that this is above the critical value for 0.05 for the signed rank test on 12 subjects, so we do not reject the null hypothesis and assume that there was no change.

An n of 12 is too small for the normal approximation to be correct, but if we substitute $T = 21$ and $n = 12$ in **Equation 12**, then we get z = 1.37 and a P value of 0.17.

If the set of differences between pairs in a paired sample are used, then this is logically the same as the test of a single sample, so we could use the sign

Table 6
Baseline and Treatment Scales and Change Ranks and Signed Ranks

Baseline	Treatment	Difference	Absolute value	Rank	Signed rank
7.81	7.88	–0.07	0.07	1	–1
4.56	0.40	4.16	4.16	12	12
8.22	9.20	–0.98	0.98	5	–5
9.14	6.29	2.85	2.85	9	9
5.41	3.34	2.07	2.07	8	8
8.81	8.12	0.69	0.69	2	2
6.99	6.14	0.85	0.85	3	3
9.86	6.99	2.86	2.86	10	10
4.68	3.04	1.65	1.65	7	7
3.45	4.40	–0.95	0.95	4	–4
5.06	8.91	–3.85	3.85	11	–11
6.63	5.23	1.40	1.40	6	6

test to test the null hypothesis that the median is zero. However, the Wilcoxon signed rank test assumes that the values are distributed symmetrically about the median, uses more information, and, in general, is more powerful than the sign test (see **Chapter 4**). We could also use the Wilcoxon signed rank test with a single sample if the assumption of symmetry about the median was reasonable.

Example 4 (Continued)

If we use the sign test to test the hypothesis of a median equal to zero, we have 8 of 12 samples above the median. This gives us a 2-sided P value of 0.3877. The large sample approximation is $z = 0.86$, which gives us a 2-sided P value of 0.3897. This result is approximately double the P value obtained from the Wilcoxon signed rank test and therefore, although neither value is in the critical region, agrees with the notion that the Wilcoxon signed rank test is more powerful when the assumptions are met.

4.3. The Wilcoxon-Mann-Whitney Test to Compare Two Groups

We next look at the test known as the *Wilcoxon rank sum test* and sometimes as the *Mann-Whitney U test*. The underlying concept is that if two populations have the same distribution, then if you mix them together you would expect to find equal numbers from both groups above and below the overall median. Using θ to represent the true shift between the 2 distributions, the null hypothesis is

$$H_0 : \theta = 0$$

$$H_1 : \theta \neq 0.$$

We begin by computing the ranks for the combined group of subjects. To use the standard notation, we assume that one group has m subjects and the other has n subjects, and m and n need not be equal.

We then compute S_m and S_n, which are the sums of the ranks of the subjects in the groups with m and n subjects, respectively. The test statistic can be calculated for either m or n:

$$\begin{aligned} U_m &= S_m - 0.5m(m+1) \\ U_n &= S_n - 0.5n(n+1). \end{aligned} \tag{13}$$

Because $U_m = mn - U_n$, only one need be calculated. For small samples, the critical region must be obtained from tables ***(4)*** or from software, and is usually based on the smallest sample. The large sample approximation (for either m or $n > 20$ or both) is

$$z = \frac{U + 0.5 - 0.5mn}{\sqrt{mn(m+n+1)/12}}, \tag{14}$$

where U is the smaller of U_m and U_n. This statistic has approximately a standard normal distribution. (Note the 12 in the denominator is fixed. It is not related to the sample size, although it may seem so in the next example.)

Example 5

Table 7 shows the depression scales for unipolar and bipolar subjects and the rank for each value using the total group, where $m = n = 12$. The U statistics from **Equations 13** are

$$U_m = 176 - 0.5 \times 12 \times 13 = 98 \quad \text{and}$$

$$U_n = 12 \times 12 - 98 = 46.$$

From tables ***(4)***, the 5% critical region for a 2-tailed test is $U \leq 37$, therefore we do not reject the null hypothesis.

The large sample statistic from **Equation 14** is

$$z = \frac{46 + 0.5 - 0.05 \times 12 \times 12}{\sqrt{12 \times 12 \times 25/12}} = -1.47.$$

The 2-sided P value for -1.47 is 0.1410, so we do not reject the null hypothesis.

Table 7
Depression Scores and Overall Ranks for 2 Groups of Subjects

	Unipolar		Bipolar	
	Score	Rank	Score	Rank
	7.81	19	8.52	21
	4.56	7	7.07	17
	8.22	20	6.34	13
	9.14	23	7.32	18
	5.41	11	5.78	12
	8.81	22	6.77	15
	6.99	16	3.68	3
	9.86	24	4.93	9
	4.68	8	3.27	1
	3.45	2	4.27	6
	5.06	10	4.19	5
	6.63	14	4.09	4
Sum of ranks		176		124

5. Comparisons of Means in More Than Two Groups: ANOVA

5.1. ANOVA

In many situations, you want to compare the means in more than 2 groups. The null hypothesis is that the means in all the groups are equal. One way to do this would be to run *t*-tests as in **Section 3.2** on all possible pairs of groups. There are several problems with this approach. First, you are ignoring the structure of the design but not looking first to see if there is any difference between any of the groups. Second, you are also ignoring the design because each *t*-test will use a measure of the variation in the data based on only the 2 groups being tested, which will differ for each test, instead of using all the data to estimate the variation in the total sample. Third, you are increasing the probability of finding a difference where none exists (type I error; **Chapter 4**). To understand it, suppose you have 3 groups so you do 3 *t*-tests. You set the critical value of α to 0.05 for each test. Then the probability of 1 or more significant results if the null hypothesis is true is $1 - (1 - 0.05)^3 \approx 0.15$. This is called the problem of multiple comparisons, and you will see it addressed further on in this chapter.

Analysis of variance was developed to analyze this type of data without creating these problems. The basic analysis is a global test for any differences in means between groups. These can be followed by secondary tests to locate these differences, using the information from the total sample and adjusting for multiple comparisons. The groups are assumed to represent some characteristic referred to as the *factor* or independent variable, and the analysis tests the *effect* of this factor on the outcome. (Sometimes the terms *factor* and *effect* are used interchangeably.) The different groups are referred to as *levels* of the factor, whether or not that is a literal description. Levels may refer to groups that represent different doses of a drug or groups which represent different diagnoses.

The logic behind ANOVA is as follows. Assume you would like to compare the means between g groups, where $g \geq 3$. Each group has an underlying normal distribution, and, very importantly, all g groups have the same underlying variance. We denote the mean of each group as $\bar{x}_i$, the standard deviation as S_i, the sample size as n_i, and the jth sample in group i as x_{ij}. Treating all samples as a single group, the total sample size is N, the grand mean is denoted $\bar{x}$, and the total variation is S_t^2. Recall that the variance is calculated as the sum of squares around the mean divided by the sample size −1. Each component of the sum of squares around the grand mean can be rewritten as follows:

$$x_{ij} - \bar{x} = (x_{ij} - \bar{x}_i) + (\bar{x}_i - \bar{x}). \tag{15}$$

Thus the total variation can be divided into 2 parts, the variation due to the difference of each sample from its group mean, and the variation due to the difference between the group means and the grand mean. If the group means are equal, then the variation due to the difference between them and the grand mean should be small; if at least 2 of the group means are different, then this component should be larger. This is illustrated in **Figures 2A** and **2B**. Both have 3 groups of measurements, each group has the same variance. The three groups are shown on the left-hand side of the figure. The right-hand side of each figure shows the 3 means compared to the overall mean. In **Figure 2A**, the means of the 3 groups are approximately equal and the dispersion of the group means is also small. In **Figure 2B**, the means are very different, and the variation of the group means from the overall mean is large.

Calculations for the ANOVA are based on the sum of squares of deviations from the mean as shown in **Table 8**. Each sum of squares has a degrees of freedom associated with it.

SSB is the *between groups sum of squares* of deviations of the group means from the overall mean. The degrees of freedom are equal to the number of groups minus 1.

SSE is the sum of squares of the deviation of each sample from its group mean, often called *the error sum of squares*. It has degrees of freedom equal to the total sample size minus the number of groups.

When the assumptions are not met:

- Non-normality & Unequal variances:
  - Kruskal-Wallis test: a non-parametric counterpart to the one way ANOVA
- Related Groups:
  - For related measures use:
  - Repeated-measures ANOVA: used when the dependent variable is measured at different points in time.

ANOVA: a special regression analysis

- **Note**:
  - Analysis of variance is a special type of regression analysis.
  - Most data sets for which analysis of variance is appropriate can be analyzed by regression with the same results.

Module 8: Comparing Groups — t-tests & Non-parametric Tests

Module Assignment

Source notes include one-sample, independent-samples and paired t-test exercises, comparison of means, and non-parametric alternatives.

1. Select an appropriate test for: mean birth weight versus 2500 g; total cholesterol in males versus females; weight before versus after treatment; and skewed hospital stay in two independent groups.

2. For each choice, state the outcome variable, grouping/paired structure, key assumption and null hypothesis.

3. Perform at least one selected test in Excel, Jamovi or SPSS.

4. Report the estimate/difference, 95% confidence interval where available, p-value and a clinical interpretation.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.